

ISC Assignment-04

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Question 1: Write a python program to encrypt and decrypt messages using the Playfair Cipher. The program should construct a 5x5 key matrix, format the plaintext into digraphs, and perform encryption and decryption.

Code:

```
# AI 331 - Information Security and Cryptography
```

```
# Lab Assignment 4
```

```
# Playfair Cipher
```

```
def make_matrix(key):
```

```
    key = key.upper().replace("J", "I")
```

```
    s = ""
```

```
    # Add key characters
```

```
    for ch in key:
```

```
        if ch.isalpha() and ch not in s:
```

```
            s += ch
```

```
# Add remaining alphabet
for ch in "ABCDEFGHIJKLMNOPQRSTUVWXYZ":
    if ch not in s:
        s += ch
```

```
# Create 5x5 matrix
mat = []
for i in range(0, 25, 5):
    mat.append(list(s[i:i+5]))

return mat
```

```
def show_matrix(mat):
    print("\n5 x 5 Key Matrix:")
    for row in mat:
        print(" ".join(row))
```

```
def format_text(text):
    text = text.upper().replace("J", "I")
```

```
# Remove spaces and non-alphabetic characters
```

```
s = ""
```

```
for ch in text:
```

```
    if ch.isalpha():
```

```
        s += ch
```

```
pairs = []
```

```
i = 0
```

```
while i < len(s):
```

```
    a = s[i]
```

```
    if i + 1 == len(s):
```

```
        pairs.append(a + "X")
```

```
        i += 1
```

```
    elif s[i] == s[i + 1]:
```

```
        pairs.append(a + "X")
```

```
        i += 1
```

```
    else:
```

```
        pairs.append(a + s[i + 1])
```

```
        i += 2
```

```
return pairs
```

```
def find_pos(mat, ch):  
    for r in range(5):  
        for c in range(5):  
            if mat[r][c] == ch:  
                return r, c
```

```
def encrypt_pair(a, b, mat):  
    r1, c1 = find_pos(mat, a)  
    r2, c2 = find_pos(mat, b)
```

```
# Same row
```

```
if r1 == r2:  
    return mat[r1][(c1 + 1) % 5] + mat[r2][(c2 + 1) % 5]
```

```
# Same column
```

```
elif c1 == c2:  
    return mat[(r1 + 1) % 5][c1] + mat[(r2 + 1) % 5][c2]
```

```
# Rectangle
```

else:

return mat[r1][c2] + mat[r2][c1]

def decrypt_pair(a, b, mat):

r1, c1 = find_pos(mat, a)

r2, c2 = find_pos(mat, b)

Same row

if r1 == r2:

return mat[r1][(c1 - 1) % 5] + mat[r2][(c2 - 1) % 5]

Same column

elif c1 == c2:

return mat[(r1 - 1) % 5][c1] + mat[(r2 - 1) % 5][c2]

Rectangle

else:

return mat[r1][c2] + mat[r2][c1]

----- MAIN PROGRAM -----

```
print("==== PLAYFAIR CIPHER ====")

key = input("Enter key: ")
text = input("Enter plaintext: ")

# 1. Create matrix
mat = make_matrix(key)

show_matrix(mat)

# 2. Format plaintext
pairs = format_text(text)

print("\nPlaintext Digraphs:")
print(" ".join(pairs))

# 3. Encrypt
cipher = ""

for p in pairs:
    cipher += encrypt_pair(p[0], p[1], mat)

print("\nCiphertext:", cipher)
```

4. Decrypt

plain = ""

for i in range(0, len(cipher), 2):

plain += decrypt_pair(cipher[i], cipher[i + 1], mat)

print("Decrypted :", plain)

Output:-

```
• PS D:\PROGRAMMING\programms\ISC> cd "d:\PROGRAMMING\programms\ISC\LAB-4" ; python Q1.py
===== PLAYFAIR CIPHER =====
Enter key: MONARCHY
Enter plaintext: INSTRUMENTS

5 x 5 Key Matrix:
M O N A R
C H Y B D
E F G I K
L P Q S T
U V W X Z

Plaintext Digraphs:
IN ST RU ME NT SX

Ciphertext: GATLMZCLRQXA
Decrypted : INSTRUMENTSX
```